This article was downloaded by: [137.68.121.64] On: 29 October 2025, At: 00:32
Publisher: Institute for Operations Research and the Management Sciences (INFORMS)
INFORMS is located in Maryland, USA

Information Systems Research

Publication details, including instructions for authors and subscription information:
http://pubsonline.informs.org

The Effect of Voice AI on Digital Commerce

Chenshuo Sun, Zijun Shi, Xiao Liu, Anindya Ghose

To cite this article:
Chenshuo Sun, Zijun Shi, Xiao Liu, Anindya Ghose (2025) The Effect of Voice AI on Digital Commerce. Information Systems Research 36(2):1147-1166. https://doi.org/10.1287/isre.2022.0140

Full terms and conditions of use: https://pubsonline.informs.org/Publications/Librarians-Portal/PubsOnLine-Terms-and-Conditions

This article may be used only for the purposes of research, teaching, and/or private study. Commercial use or systematic downloading (by robots or other automatic processes) is prohibited without explicit Publisher approval, unless otherwise noted. For more information, contact permissions@informs.org.

The Publisher does not warrant or guarantee the article's accuracy, completeness, merchantability, fitness for a particular purpose, or non-infringement. Descriptions of, or references to, products or publications, or inclusion of an advertisement in this article, neither constitutes nor implies a guarantee, endorsement, or support of claims made of that product, publication, or service.

Copyright © 2025, INFORMS

Please scroll down for article—it is on subsequent pages

With 12,500 members from nearly 90 countries, INFORMS is the largest international association of operations research (O.R.) and analytics professionals and students. INFORMS provides unique networking and learning opportunities for individual professionals, and organizations of all types and sizes, to better understand and use O.R. and analytics tools and methods to transform strategic visions and achieve better outcomes.
For more information on INFORMS, its publications, membership, or meetings visit http://www.informs.org

INFORMATION SYSTEMS RESEARCH
Vol. 36, No. 2, June 2025, pp. 1147–1166
ISSN 1047-7047 (print), ISSN 1526-5536 (online)
https://pubsonline.informs.org/journal/isre

Downloaded from informs.org by [137.68.121.64] on 29 October 2025, at 00:32 . For personal use only, all rights reserved.

The Effect of Voice AI on Digital Commerce

Chenshuo Sun,[a,b,c,*] Zijun Shi,[d] Xiao Liu,[e] Anindya Ghose[e]

[a] School of Business, Renmin University of China, Beijing 100872, China; [b] Department of Information Systems and Analytics, National University of Singapore, Singapore 117418; [c] College of Business, City University of Hong Kong, Hong Kong; [d] Department of Marketing, Hong Kong University of Science and Technology, Clear Water Bay, Hong Kong SAR, China; [e] New York University, New York, New York 10012

*Corresponding author

Contact: csun@stern.nyu.edu, https://orcid.org/0000-0001-7708-2671 (CS); juneshi@ust.hk (ZS); xl23@stern.nyu.edu, https://orcid.org/0000-0002-7093-8534 (XL); aghose@stern.nyu.edu, https://orcid.org/0000-0002-6499-8944 (AG)

Received: February 18, 2022
Revised: February 22, 2023; November 29, 2023; April 30, 2024
Accepted: August 10, 2024
Published Online in Articles in Advance: May 6, 2025

https://doi.org/10.1287/isre.2022.0140

Copyright: © 2025 INFORMS

Abstract. Voice-activated shopping assistants (voice AI), such as Amazon Alexa and Alibaba Tmall Genie, have been gaining popularity worldwide as a new channel for online shopping. In this paper, we analyze large-scale archival data of consumer-level purchase records from Alibaba, the world's largest e-commerce platform, to empirically investigate how consumers' adoption of Tmall Genie affects their consumption. The results show that the average consumer's weekly spending the e-commerce platform increased by 16.6% within the first four months after adopting voice AI. Additionally, we explore specific product features that moderate the effect of Genie adoption by examining repeat purchases, product substitutability, and familiarity, supporting a mechanism that involves reducing information acquisition costs. The positive effects of Genie adoption remain significant on repeat purchases in the long term, although they attenuate over time. Furthermore, our analyses reveal that on average, the voice channel has a positive spillover effect on spending on the PC channel but no significant effect on the mobile channel. The channel dynamics are contingent on specific shopping contexts. Our results demonstrate that voice AI devices with shopping capabilities can enhance the growth of the affiliated e-commerce platform. As the first study to empirically examine the impact of voice AI adoption on e-commerce consumption, our paper provides valuable implications for e-commerce platforms and retailers leveraging voice-activated shopping.

History: Deepa Mani, Senior Editor; Guodong (Gordon) Gao, Associate Editor.
Funding: The authors acknowledge the research grant and support from the Marketing Science Institute (MSI). C. Sun acknowledges the support from National University of Singapore under the Start-Up Grant Scheme and the AI & Management (AIM) Research Center at Tsinghua SEM.
Supplemental Material: The online appendix is available at https://doi.org/10.1287/isre.2022.0140.

Keywords: voice AI • voice shopping • e-commerce • causal inference • heterogeneity • channel dynamics • transactional cost

1. Introduction

In recent times, generative AI, notably exemplified by ChatGPT and DeepSeek, has garnered pronounced attention for its remarkable capabilities, reflecting transformative potentials in creating humanlike texts and information (Brynjolfsson et al. 2023, Goli and Singh 2024, Hu et al. 2024, Li et al. 2024, Yeverechyahu et al. 2024, Zhang et al. 2024). The expanding utilization of conversational AI in market research and e-commerce facilitates a nuanced understanding of consumer preferences and scaling the AI-enabled e-commerce market. This market is expected to reach $16.8 billion by 2030, underscoring its pivotal role in deciphering and potentially reshaping consumer behaviors and market research (Forbes 2023, Li et al. 2024). Voice-activated smart devices, backed by conversational AI, are rapidly gaining popularity among companies and consumers. In the United States, Amazon Alexa, Google Home, and Apple Homepod have become household names, and in China, Tmall Genie, Baidu Xiaodu, and Microsoft Xiaoice are widely adopted.[1]

Coupled with the rise of these devices is the burgeoning trend of voice-activated shopping, where consumers interact with a voice AI interface to place orders. For instance, shopping through Amazon Alexa saw a threefold increase between the 2017 and 2018 holiday seasons. Walmart, recognizing this trend, partnered with Google in April 2019 to launch Walmart Voice Order, enabling Google Home users to order directly from Walmart's online portal using voice commands. Surveys done by Capital One Shopping show that in 2024, 49% of U.S. consumers use voice search for shopping.[2]

Despite the significance of voice shopping in consumers' daily lives and the digital economy, little is known about how the adoption of a voice-activated shopping device impacts a consumer's subsequent shopping behaviors. Voice shopping could plausibly affect a consumer's purchase journey. For example, if a consumer is browsing videos from a popular cosmetics blogger and encounters a video about lipsticks, the consumer can immediately start the search process by

Downloaded from informs.org by [137.68.121.64] on 29 October 2025, at 00:32 . For personal use only, all rights reserved.

asking, "Alexa, what are the best-selling lipsticks?" If Alexa generates a desirable suggestion, the consumer can place an order in seconds by saying, "Add to my cart and check out."[3] In this scenario, consumers start with an "I want-to-know moment" conversation with voice AI, followed by an "I want-to-buy moment," which can be accomplished with the help of voice AI.

The potential impact of voice AI device adoption on consumer spending remains ambiguous and warrants rigorous empirical examination. Two contradictory effects could be at work simultaneously. On one hand, voice AI devices enable consumers to shop hands-free;[4] the vocal search process can make it convenient for consumers to efficiently locate specific products while multitasking, potentially leading to increased spending.[5] Additionally, the ease of voice shopping might result in more impulsive purchases during interactions with the voice AI.[6] On the other hand, research on auditory consumer behavior indicates that information presented by voice could be more difficult to process than information presented in text or visually (Munz 2020). Furthermore, the perception of invasiveness associated with voice AI may also deter adoption from consumers that prioritize retaining control over their data (Mayya and Viswanathan 2024). Compounding this, inaccuracies in voice AI recognition not only cause inconvenience but also result in significant time losses, as consumers are often forced to restart the search process with increased forethought (Melumad 2023). This increased cognitive effort required for spoken communication could discourage thorough product exploration and, subsequently, reduce purchasing intentions. These opposing predictions create a tension that underscores the need for a comprehensive empirical study that identifies the causal effects of voice AI adoption on e-commerce consumption. Our research is thus crucial not only for advancing academic understanding but also for guiding e-commerce platforms leveraging voice AI.

To the best of our knowledge, our study is the first empirical examination of this topic, and therefore, this paper contributes to the growing field of research on the impact of emerging technologies on consumption (Ghose 2018, Adamopoulos et al. 2020, Pamuru et al. 2021, Xu et al. 2023) and studies on voice AI and chatbots (Luo et al. 2019, Ba et al. 2020, Smith 2020, Luo et al. 2021, Bergner et al. 2023, Melumad 2023, Son et al. 2023). Beyond examining the overall impact of voice AI adoption on consumption, this paper also seeks to understand the categories of products that are most affected by voice AI adoption. Previous studies have demonstrated that the impact of technology adoption varies across products, depending on factors such as the time and effort required for product evaluation (Adamopoulos et al. 2020, Park et al. 2020). By delving into the heterogeneous effects of voice AI adoption on spending across different product categories, we provide deeper insights into how product features associated with the cost of information acquisition might moderate the main effect, which can assist businesses in efficiently leveraging voice AI for their marketing strategies. In addition, the impact of voice AI adoption could change over time, leaving the long-term stability of its effect on spending an unclear question. In this paper, we deepen the understanding of the effectiveness of voice AI on spending by conducting additional analyses to examine the temporal dynamic effects of voice AI adoption. Furthermore, today's consumers have access to various digital shopping channels, including mobile, PC, and tablet, allowing them to choose the most suitable one for their specific needs and preferences. Prior literature has also examined the spillover effects of the adoption of one shopping channel on other channels (Ghose et al. 2013, Xu et al. 2017). While voice AI adoption could enhance overall consumption by making it more intuitive and engaging, it is also possible that consumers might shift their shopping activities from mobile and PC channels to the voice channel. Due to different comparative advantages among different channels, the spillover effects on specific channels will be contingent upon customer preferences and channel-specific characteristics. Consequently, the influence of adopting voice AI on the other two channels may depend on specific shopping contexts. The impact of voice AI adoption on the PC or mobile channel remains uncertain, necessitating empirical investigation for a comprehensive understanding. The findings of these heterogeneity analyses bolster the overall contribution of the paper because they further empirically verify the underlying mechanisms.

We leverage a large-scale archival data set from Alibaba, one of the world's largest online retailers. Alibaba developed and launched Tmall Genie (or "Genie" for short), its own voice AI device similar to Amazon Alexa, offering a voice shopping experience. Our unique transactional data set contains 31 weeks of consumer-level purchase records, including both Genie adopters and nonadopters. We examine how the adoption of Genie as a shopping channel affects consumer expenditure on the focal platform. We employ granular propensity score matching (PSM) to match adopters with nonadopters and then use the difference-in-differences (DiD) approach to identify the effect of Genie adoption on spending.

Our analyses yield several important findings. First, we find that the adoption of Tmall Genie, the voice AI device launched by the e-commerce platform, leads consumers in our sample to increase their average weekly spending on Tmall by 16.6% within the first four months after adoption. Second, we test how product characteristics moderate the effect of Genie adoption, and we provide evidence that the mechanism involves reduction of information acquisition costs. Third, our findings indicate a stronger Genie adoption

Downloaded from informs.org by [137.68.121.64] on 29 October 2025, at 00:32 . For personal use only, all rights reserved.

impact on spending in product categories that require less active search or comparison, specifically, product categories with low substitutability (e.g., books) and product categories that the focal consumer makes frequent purchases in and is more familiar with. This finding bolsters the overall contribution of the paper, as it further empirically verifies the underlying mechanism. Fourth, we find that the overall treatment effect of Genie adoption on spending attenuates over time. For repeat purchases in the voice channel, although the positive effect weakens over time, it remains statistically significant in a relatively long term after adoption.

Moreover, we demonstrate a significantly positive spillover effect of Genie adoption on the PC channel while observing no significant impact on the mobile channel across all the product categories. Further category-level investigation reveals the nuanced dynamics within the PC channel, wherein a positive spillover to the purchase of unfamiliar items and a negative spillover to the purchase of familiar items are observed. Within the mobile channel, voice AI leads to more purchases stemming from casual conversations with Genie, whereas it crowds out hands-free purchases that can be done with Genie. These findings also inform, among others, future literature on the interdependencies among shopping channels. Lastly, from a social equity perspective, we find that older consumers are equally capable of benefiting from voice AI and may be inclined toward voice-based interaction over mobile devices. Additionally, we observe that the positive spillover effects on spending are primarily because of an increase in purchase quantity rather than higher prices. Overall, we demonstrate that voice-activated shopping devices can be well positioned to enhance the growth of the affiliated e-commerce platform.

The rest of this paper is organized as follows. In Section 2, we provide an overview of the relevant literature and formalize the rationale and hypotheses behind the empirical analyses. Section 3 details the institutional background and the data. Section 4 presents the identification strategies as well as the main empirical results. We present the methods and results in testing the remaining hypotheses in Sections 5, 6, and 7, respectively. Section 8 discusses the results from post hoc analyses and extensive robustness checks. Finally, Section 9 concludes with the implications of our study.

2. Literature and Hypothesis Development

Our work contributes to several streams of relevant research, including how the adoption of digital technologies affects consumption behavior, the implications of voice AI technology, as well as the channel dynamics.

2.1. Related Literature

2.1.1. Digital Technology Adoption and Consumption. Existing literature has focused on understanding the implications of emerging digital technologies on consumer behavior and service providers. On the consumer side, for instance, Xu et al. (2023) investigated the determinants and effects of mobile payment technology adoption on consumer consumption behaviors. On the service provider side, the adoption of emerging technologies can open up new revenue streams. Pamuru et al. (2021) showed that restaurants incorporating augmented reality (AR) experiences benefit from enhanced consumer engagement and improved. Tan and Netessine (2020) studied the impact of tabletop technology on restaurant sales performance and found a great potential for introducing this technology in a large service industry that lacks digitalization. Combined, such new technologies create new avenues for consumers to engage with services, potentially altering their purchasing decisions and the outcomes for the service providers. The introduction of the mobile channel (Ghose 2018, Park et al. 2020) and the internet of things (IoT) (Adamopoulos et al. 2020) have been shown to be able to reshape consumer behavior and influence product sales in their respective ways. Overall, the use of emerging technologies has the potential to significantly impact the consumer-provider relationship, altering the way services are delivered and consumed and how consumers interact with them. Voice AI differs from the technologies discussed in existing studies, primarily because of its unique voice modality that alters consumer behaviors in searching and purchasing (Melumad 2023, Son et al. 2023). This divergence highlights the need for further empirical investigation. Although voice AI gains attraction across various industries, the effects of such voice-enabled technology on e-commerce consumption remain unclear. By comprehensively examining the adoption and utilization of this novel voice AI technology, our study extends the literature on the impact of emerging technologies on consumption.

2.1.2. Contemporary Literature on Voice AI. Moreover, this work contributes to the expanding body of literature on the impact of voice AI, specifically in the context of e-commerce. With the escalating use of voice AI technology, understanding the impact of voice AI adoption on consumer behavior becomes increasingly vital. Despite that, we know little about how voice assistants, as a new channel for online shopping, affect e-commerce consumption. Prior literature has elaborated on the intelligent agent technology applications and adoption via a theoretical framework (Kumar et al. 2016), and it has demonstrated the difficulty of processing auditory information in voice commerce using experiments (Munz 2020). Moreover, researchers have also conducted case studies, surveys, and algorithmic designs to investigate the implications and opportunities of voice AI technology in marketing and sales policies (Ba et al. 2020, Smith 2020, Chen et al. 2023).

Downloaded from informs.org by [137.68.121.64] on 29 October 2025, at 00:32 . For personal use only, all rights reserved.

Additionally, Luo et al. (2019) analyzed field experiment data to study how revealing a chatbot's identity negatively affects customer purchase decisions. Melumad (2023) studied the effects of voice technology on online information searches and found that vocalizing queries resulted in more specific and detailed descriptions, as evidenced by survey data. Existing research on voice AI predominantly takes a descriptive, theoretical, survey, or qualitative approach. Although Son et al. (2023) demonstrated the effect of smart speaker usage on consumers' digital content (namely, video on demand) searches and purchases, empirical evidence remains scarce regarding how the adoption of voice AI influences consumption on e-commerce platforms, a crucial area where voice AI is expected to play a significant role. This research aims to bridge this gap. To the best of our knowledge, this paper is among the first to empirically study the impacts of adopting voice AI on consumers' e-commerce consumption.

2.1.3. Channel Dynamics. The advent of the internet and the rise of emerging technologies have broadened the array of channel options available to consumers for making purchases (Sun et al. 2022). Existing research in information systems and marketing has investigated the interdependencies between various shopping channels and has demonstrated that the impact of new channel introduction can vary depending on the unique characteristics of the channels. For instance, Ghose et al. (2013) explored how consumers' online browsing behaviors vary between personal computers and mobile channels and showed that the smaller screen size of mobile phones increases the cost of browsing. Bang et al. (2013) found that the impacts of mobile channel introduction on traditional online channels depend on the fit between channel capabilities and product characteristics. In this regard, the relevant stream of literature has revealed that adopting additional shopping channels can have nuanced and multifaceted effects, depending on the specific capabilities of these channels.

Previous research has shown that adopting a new channel can lead to a complementary effect, enhancing the utility of existing channels, and a cannibalization effect, reducing the usage of existing channels. For instance, Xu et al. (2014) found a significant increase in demand at the corresponding mobile news site in response to the introduction of a mobile channel. Xu et al. (2017) documented that tablet adoption substitutes the PC channel but complements the smartphone channel. Likewise, Xu et al. (2023) found that the mobile payment channel acted as a substitute for the offline channel and as a complement for the PC channel, and both effects increased over time. Despite the proliferation of studies on channel dynamics, there is no definitive tone on whether voice AI will hold a substitute or complementary relationship with the existing channels. Our work contributes to this strand of literature by evaluating how voice assistants, as a new shopping channel, affect traditional PC and mobile purchase channels.

2.2. Hypotheses Development

In this section, we develop four hypotheses focusing on the main effect, the moderating role of product features, dynamic effects, and channel spillover effects.

Extant literature has documented the impact of emerging technologies on consumption behavior. For example, the introduction of the mobile channel for news (Xu et al. 2014) and mobile payment (Xu et al. 2023) has substantially increased demand and consumption. The adoption of mobile shopping has been found to increase search activities and sales of leading products (Park et al. 2020). Mobile devices such as smartphones and tablets, offering convenient access, encourage consumers to incorporate mobile shopping into their habitual routines. Xu et al. (2017) found that the introduction of tablets can spur casual browsing, leading to more impulsive and diverse product purchases, thereby enhancing overall e-commerce sales. In a similar vein, the internet of things has also demonstrated a significant increase in sales because of its ability to enhance convenience and reduce intangible transaction costs for consumers in purchase processes (Adamopoulos et al. 2020). As voice AI technology develops and becomes more prevalent in a range of industries, it is likely to have a similar impact on consumption and spending. Voice AI, enabling hands-free shopping, offers greater convenience and flexibility. The enhanced user experience could lead to increased usage and higher spending on some particular product categories from consumers. Overall, we expect the adoption of voice AI will have a positive overall impact on consumer spending on the focal platform. Based on these considerations, we propose the following baseline hypothesis:

Hypothesis 1. *Voice AI adoption has a positive overall effect on consumer spending.*

Despite the wide adoption of voice-enabled agents for their convenience, several attributes of the voice shopping channel can affect the types of products in which the increase in consumer spending is reflected. Research on auditory consumer behavior indicates that information presented through voice is generally more difficult to process than information presented through text or visuals (Munz 2020). Given the perceived higher cognitive cost of searching and comparing unfamiliar products through voice AI, consumers may be deterred from exploring a wider range of options and instead only focus on familiar products. Consumers also have different attitudes toward convenience goods and other shopping goods, with convenience goods being those

Downloaded from informs.org by [137.68.121.64] on 29 October 2025, at 00:32 . For personal use only, all rights reserved.

that are purchased repeatedly and require minimal search effort, whereas others require a more serious process of selection (Benabou 1993). Park et al. (2020) showed that for convenience goods, which are low in search needs because of repeated purchases and enough prior knowledge, the adoption of mobile channels led consumers to choose the mobile channel as the primary means for their searches and purchases. Similarly, Adamopoulos et al. (2020) demonstrated that the introduction of IoT technologies as a sales channel is more effective for less substitutable products, which are goods that are less differentiated and require less time and effort from consumers to thoroughly evaluate their options, validating that consumers value convenience and efficiency more when shopping for these types of products. Consequently, consumers are likely to prefer using Genie for goods that they purchase repeatedly, as it saves the time and effort needed for reordering these items using voice commands rather than having to search and compare. Hence, we anticipate Genie to be better suited for repeat purchases, familiar product categories, and those with low substitutability, where the cost of information processing is lower for consumers. We thus propose:

Hypothesis 2. *The positive effect of voice AI adoption on spending through the voice shopping channel is stronger for products that do not bear the high cost of information acquisition, including repeat purchase, more familiar, and less substitutable product categories.*

It is also worth assessing whether the positive effect of voice AI adoption on consumption remains stable over time. This could provide insight into the factors that may influence consumer spending decisions when using voice shopping. For instance, in the early stages of adoption, consumers may be hesitant in their usage of Genie because of unfamiliarity or skepticism, but as they become more familiar with it and see its benefits, they may become more inclined to use it, leading to a positive effect that increases over time. However, if consumers view voice AI as too complex or lacking, the positive effect on consumption could diminish over time. Additionally, as people become more accustomed to the technology, the novelty effect may wear off, causing the positive effect on consumption to level off.

Novelty, which can be thought of as an innate human preference for something new, unique, and different from the familiar, can be created by the freshness and innovation of information technology (Massetti 1996, Huang 2003). Manchanda et al. (2015) described the novelty effect as a temporary positive response to the adoption of something new, but this response tends to decline over time as people revert to their normal purchase behavior. Previous research has found that the novelty of the technology and its ability to perform its intended services can initially be satisfying, but as the novelty wears off, consumer expectations may increase, leading to a decrease in satisfaction with the technology (Meuter et al. 2000). When consumers are disappointed with the technology's perceived usefulness and compatibility with their everyday activities, their interest and utilization of the service will decrease (Parthasarathy and Bhattacherjee 1998). This diminishing engagement has been evident in past studies on the effects of new technology adoption in education, where the novelty effect decreases as students get familiar with the technologies, resulting in diminished motivation to use the technology. Early studies have also demonstrated that the novelty effect of robots can quickly wear off, and people have fewer interactions with them over time (Adamopoulos et al. 2020). Therefore, as the novelty of voice AI technology naturally wanes over time, it may lead to a decrease in its perceived usefulness and a corresponding decline in consumers' motivation and usage. The adoption of a new technology can be a complex process, and the impact of Genie adoption may be influenced by various factors that can change over time. In this context, we reason that the ease of use and conversational nature of voice AI Genie may make the novelty effect more significant in determining the temporal dynamics of the treatment effect. Hence, we anticipate that the positive effect of Genie adoption will attenuate over time, leading to the following hypothesis:

Hypothesis 3. *The positive effect of voice AI adoption diminishes over time.*

As Genie provides an additional channel for consumer interaction, adopters' shopping behavior through other channels may change. It is thus important to evaluate how the other shopping channels (namely, PC and mobile) are impacted, if any. Xu et al. (2017) found that tablet adoption substitutes desktop commerce but complements the smartphone channel. Though tablets share functional similarities with PCs, tablets offer a more portable and intuitive browsing experience, thereby substituting both browsing and purchase activities away from the PC channel. Meanwhile, because smartphones' smaller sizes and ubiquitous connectivity augment the shortcomings of tablets, which are more location restrictive and bulkier, users make more purchases on smartphones with fewer page browses after tablet adoption.

Likewise, as different channels have different comparative advantages, the specific channel spillover impacts will depend on customer preferences and channel-specific traits. While mobile devices have several advantages such as portability and accessibility, which can help consumers fulfill their transactions more efficiently (Kleijnen et al. 2007), the PC channel makes it easier for consumers to search and browse

Downloaded from informs.org by [137.68.121.64] on 29 October 2025, at 00:32 . For personal use only, all rights reserved.

products, as computers' large screens can display more product information (Ghose et al. 2013). Given the comparative advantages in different contexts, the impact of voice AI on mobile channels could go either way. In the case where the adopters want to make a decision after a casual conversation with voice AI but are constrained by the high cost of searching and comparing products using the voice channel, they would direct themselves to mobile devices to complete the purchase process (Xu et al. 2017). On the other hand, voice AI allows hands-free shopping (Bahmani et al. 2022), where adopters can make purchases while multitasking on mobile phones, producing a negative spillover on the mobile channel.

The spillover effect on the PC channel can also be mixed. On the one hand, consumers may switch to PC from earlier voice-initiated sessions to continue their shopping because of its expanded visual experience (Xu et al. 2017). On the other hand, for familiar items, consumers do not need much search and comparison but, instead, can easily use voice AI to complete the transaction, resulting in a cannibalization (negative) effect on the PC channel (Park et al. 2020). Thus, the impact of voice AI adoption on the other two channels should depend on the specific shopping context. Consequently, the overall net effect of the Genie adoption on the PC or mobile channel remains unclear and calls for an empirical examination. Therefore,

Hypothesis 4. *The spillover effects of Voice AI adoption on spending in the PC and mobile channels can be either positive or negative, depending on the shopping context (e.g., the product category).*

3. Institutional Background and Data Description

We acquired access to a large archival data set of consumer-level purchase data from Tmall, the online business-to-consumer retail site of the world's largest e-commerce platform, Alibaba. Since its inception in April 2008, Tmall has acquired more than 726 million registered users. In 2019, Tmall's sales revenue exceeded Chinese yuan (RMB) 5.7 trillion (USD $853 billion),[7] accounting for 55.9% of online retail sales in China.[8] Traditional online shopping channels at Tmall include PC and mobile. In July 2017, Alibaba launched Tmall Genie, an AI-powered smart device, to link customers with new shopping experiences, smart home appliances, and other new services. As with similar voice assistant products developed by other tech giants, Genie[9] enables consumers to use voice commands to make purchases. More than one million orders were placed and paid via Genie's voice-shopping feature during Alibaba's "11.11" Global Shopping Festival in 2019.[10] Based on the early success of Genie, Alibaba announced a plan to invest RMB 10 billion (US $1.4 billion) in the AI system for voice assistants in May 2020.[11]

A typical conversation between a consumer and Genie in the voice shopping context begins with the customer's inquiry. In the illustrative example below, a consumer wants to buy a mobile refill card and does so through an efficient conversation with Genie:[12]

Customer: "Tmall Genie, I'd like to order a mobile refill card."

Genie: "I would recommend China Mobile's refill card. The total price is 100 RMB. It will be delivered to (address). May I place the order for you?"

Customer: "Yes, please place the order."

Genie: "Sure! In order to proceed, let us do voice authentication first. Please keep quiet around, and after the 'beep,' say 'Tmall Genie, 2065.'" (Here, 2065 is the authentication code randomly generated by the system.)

Customer: "2065."

Genie: "If you want to know the delivery status, you can let me know by saying 'Tmall Genie, tracking information.'"

We collaborate with Alibaba to collect data of 38,179 randomly selected consumers. Our data consist of 5.59 million consumer-level transactions from April 1 to October 31, 2019; 5.23 million purchases were made on mobile devices (including smartphones and tablets), whereas 0.35 million were made on PCs, and 0.005 million were made with voice shopping channels.[13] The purchase transaction data includes an anonymized user ID, product category name (Tmall offers 137 product categories; see Online Appendix A4), purchase date, confirmed payment amount, the shopping channel (PC, mobile, or voice), the consumer's age and gender, and the consumer's number of historical purchases prior to the first day of data collection (April 1, 2019).

3.1. Sample of Genie Adopters and Nonadopters

In 2019, Alibaba reported that over 10 million Genie units had been activated, rendering Genie the most popular voice assistant product in China.[14] With the above-described data, we leverage a small set of Genie adopters to conduct the analysis. In our sample, "adopters" are customers who adopted Genie during the first week of July (July 1–7), 2019, whereas "nonadopters" are customers who never adopted Genie at any time before the end of the sample period (October 31, 2019). Our data set has 693 adopters whose first adoption of Genie took place in the first week of July (i.e., from July 1 to July 7, 2019), a similar procedure adopted by Xu et al. (2023).[15] The remaining 37,486 control users were randomly drawn from all the users (based on their user IDs) who had not adopted Genie at any time before the end of the sample period (October 31, 2019). We place no other requirements (e.g., the number of purchases exceeds a specific threshold) for consumers to be selected into our sample, ensuring no sample selection bias in our data

Downloaded from informs.org by [137.68.121.64] on 29 October 2025, at 00:32 . For personal use only, all rights reserved.

construction that may undermine the external validity of our findings. In the posttreatment period (i.e., after the first week of July), adopters could make purchases on Tmall via all three channels (PC, mobile, and voice), whereas nonadopters were restricted to the PC and mobile channels only.

One may wonder whether some consumers chose to adopt Genie because of a nonrandom shock such as a promotion event. To address this concern, we confirmed with Alibaba that no promotion activities occurred during the first week of July 2019.[16] However, the identification of the impact of a consumer's Genie adoption on the consumer's Tmall spending remains challenging because self-selection creates endogeneity concerns. For instance, if adopters have more disposable income than nonadopters, then adopters might be expected to make more purchases on Tmall; the confounding of disposable income would lead to an overestimation of the treatment effect. We employ various econometric techniques to address the potential endogeneity concerns in the following sections.

3.2. Measuring Spending

We measure consumer-level spending as the *total amount spent on Tmall* (the confirmed payment amount across all 137 product categories) and the *amount spent in each product category*. We exclude the purchase of Genie itself from the calculations; otherwise, the outcome variables for adopters and nonadopters would not be comparable in the first place. We aggregate the raw data at the user-week level, reasoning that spending would fluctuate too much at the user-day level.

Our data collection period (April 1, 2019 to October 31, 2019) corresponds to 31 weeks (i.e., from week 14 through week 44 of 2019), so the panel data set contains 1,183,549 observations from 38,179 consumers. Of these 31 weeks, the first 13 weeks (April 1 to June 30, 2019) constitute the pretreatment period, 1 week (July 1 to July 7, 2019) constitutes the treatment period, and 17 weeks (July 8 to October 31, 2019) constitute the posttreatment period.

We report the summary statistics in Table 1. For clarity in presentation, we report statistics solely for the top 13 product categories, aggregating spending in the remaining 124 categories under the "*all others*" category.

4. Main Effect

4.1. Identification Strategy

Our goal is to estimate the average treatment effect on the treated (ATT), namely, the effect of a consumer's Genie adoption on their spending on Tmall. A major challenge we face is the lack of a randomized assignment of consumers into treatment and control groups. We use two main strategies to address this challenge. First, in the main text, we estimate a difference-in-differences model, a widely adopted approach that eliminates persistent linear and additive individual-specific effects that may introduce endogeneity. To construct an appropriate sample for the DiD estimation, we attempt to control for the unobserved need for spending in the absence of Genie by conditioning the sample on a rich set of observed characteristics (for a similar approach, see Bronnenberg et al. 2010 and Datta et al. 2018). Specifically, we use a quasiexperimental

Table 1. Summary Statistics

Variable	*N*	Mean	SD	Min	Max
User demographics					
Age	38,179	31.89	7.656	18	57
Gender (male = 1)	38,179	0.583	0.493	0	1
Number of Historical Purchases	38,179	301.2	461.8	0	13,641
Spending per category					
Spending_Mobile Refill Card	1,183,549	11.51	39.52	0	2,994
Spending_Transportation	1,183,549	2.803	54.89	0	9,368
Spending_All others	1,183,549	1.853	55.46	0	24,000
Spending_Movies & Sports Ticket	1,183,549	1.220	11.92	0	1,656
Spending_Fresh Hema	1,183,549	1.152	15.47	0	2,522
Spending_O2O Store	1,183,549	1.032	31.83	0	6,068
Spending_Hotel	1,183,549	0.414	18.30	0	6,042
Spending_Game Time Cards	1,183,549	0.285	20.41	0	7,001
Spending_Used Good	1,183,549	0.185	14.32	0	4,300
Spending_Tencent QQ	1,183,549	0.147	13.10	0	8,645
Spending_Mobile Phone Plans	1,183,549	0.087	3.734	0	1,980
Spending_Video Game	1,183,549	0.066	9.053	0	5,005
Spending_Insurance	1,183,549	0.035	3.084	0	1,435
Spending_Coffee & Beverages	1,183,549	0.015	0.998	0	294
Browsing_SmartDevices	1,183,549	0.080	2.474	0	832

Notes. Summary statistics are calculated on the unmatched sample of 693 adopters and 37,486 nonadopters, observed over 31 weeks from April 1 through October 31, 2019. The unit of analysis is the individual user for demographics and the user-week for the purchase variables. Throughout the paper, we use four significant figures (e.g., 1.000, 10.00, and 100.0) to report the results.

Downloaded from informs.org by [137.68.121.64] on 29 October 2025, at 00:32 . For personal use only, all rights reserved.

matching procedure (detailed in Section 4.3) in which we match adopters with similar nonadopters based on each user's propensity to adopt Genie, as estimated from the user's demographics and granular shopping behaviors. Moreover, consumers inclined to purchase Tmall Genie may also tend to spend more time examining smart device product pages, potentially leading to a reverse causality issue. To rule out this potential endogeneity concern, we introduce an additional matching covariate—the browsing history of smart devices—adding an additional dimension to pair customers from both treatment and control groups. Holding all other observable characteristics equal, individuals with a similar number of pageviews on the smart device product page are more likely to have the same level of interest in purchasing it.[17] Additionally, we incorporate individual and time fixed effects in the panel regressions to control for unobservable factors, and we undertake a series of checks to ensure the robustness of our findings. Moreover, in Online Appendix A13, we strengthen the main finding with an instrumental variable (IV) approach.

4.2. Comparison of Adopters and Nonadopters

First, we compare the demographics and purchasing behaviors of adopters and nonadopters during the pretreatment period. Table 2, panel A shows that, on average, adopters and nonadopters differ significantly: adopters are younger than nonadopters (31.0 years old versus 31.9 years old, $p = 0.002$), and the adopter group has a smaller share of males (39.0% versus 58.7%, $p < 0.001$). Furthermore, in the pretreatment period, adopters spent more per week than nonadopters (RMB 34.35 versus RMB 19.36, $p < 0.001$) and had more historical purchases (560.0 versus 296.4, $p < 0.001$). Regarding the browsing history of smart devices, adopters have a higher number of pageviews per week compared with nonadopters during the pretreatment period (0.324 versus 0.076, $p < 0.001$). These significant pretreatment differences between the treatment and control groups might explain some or all of the differences in spending in the posttreatment period, thus posing a challenge to the clean identification of Genie adoption's effects. In the next section, we use quasiexperimental methods to address self-selection.

4.3. Propensity Score Estimation and Matching

To mitigate significant differences between the treatment and control groups and thereby alleviate the self-selection issue, we match each adopter with nonadopters who have a similar propensity of adoption, based on similarities in observable characteristics (while including individual fixed effects could partially address the selection issue, this approach imposes a linear functional form, whereas matching allows for more flexibility). Our matching utilizes one-to-five matching with replacement to derive the closest matched nonadopters. As a result, each adopter is paired with nonadopters who have similar propensities of being treated. This approach enables a fair comparison of shopping behaviors between these two groups of consumers. In particular, we estimate each individual's adoption propensity as a function of observed variables

$$Pr\,(Adoption_i) = Pr\,(\varphi_0 + \boldsymbol{\varphi}^T \boldsymbol{Z}_i + \epsilon_i > 0), \qquad (1)$$

where $\boldsymbol{Z}_i$ is a vector of observed individual-specific characteristics, $\boldsymbol{\varphi}$ is a vector of coefficients, φ_0 represents the constant term, and ϵ_i represents the error term. We assume that ϵ_i are independent and identically distributed random variables with a type I extreme value distribution, making the propensity score estimation a logit model.

The covariates in $\boldsymbol{Z}_i$ describe the user's demographics (age and gender) and pretreatment purchase behavior (the number of historical purchases, spending in each product category, and browsing history of the smart devices). Note that we included highly granular purchase behavior data (at the product category level) to ensure that the matched sample contains adopters and nonadopters whose purchase behaviors are as similar as possible. However, not all of the 137 product categories defined by Alibaba had sufficient nonzero observations for matching,[18] so we used the 13 categories with the highest purchase frequency and aggregated the remaining 124 categories (labeled "*all others*"), as shown in Table 2. Thus, we used consumer purchase behaviors in 14 categories to conduct matching.[19] Figure 1 shows the propensity score distribution for the treatment and control groups before and after matching. The score distributions become much more similar after matching, indicating that the two groups are well matched and covariate imbalance has been greatly reduced (see Table 2, Panel B).

4.4. Main Effect Using Difference-in-Differences

We estimate the difference-in-differences model on the matched sample to quantify the effect of a consumer's Genie adoption on the consumer's spending on the focal e-commerce platform. We estimate the main effect using the following equation:

$$y_{it} = \gamma \cdot (Adoption_i \times Post_t) + \delta_i + \tau_t + \varepsilon_{it}, \qquad (2)$$

where y_{it} is the total amount spent by consumer i in week t, γ captures the treatment effect, δ_i is consumer-level fixed effects, τ_t is week-level fixed effects, and ε_{it} is the random error term. We report robust standard errors clustered at the user level. In summary, we combine the propensity score matching method (to select nonadopters who are like adopters) and the DiD approach. We assume that the unobserved ϵ_i in the

Downloaded from informs.org by [137.68.121.64] on 29 October 2025, at 00:32 . For personal use only, all rights reserved.

Table 2. Comparison of Adopters and Nonadopters Before and After Matching

Variables	Adopter (*N*)	Adopter (mean)	Nonadopter (*N*)	Nonadopter (mean)	*p*-value of *t*-test
		Panel A: Before matching			
Age	693	31.00	37,486	31.90	0.002
Gender (male = 1)	693	0.390	37,486	0.587	0.000
# Historical Purchases	693	560.0	37,486	296.4	0.000
Spending_Mobile Refill Card	693	17.00	37,486	11.52	0.000
Spending_All others	693	3.722	37,486	1.963	0.004
Spending_Transportation	693	2.520	37,486	1.448	0.033
Spending_Movies & Sports Ticket	693	2.348	37,486	0.992	0.000
Spending_Fresh Hema	693	2.123	37,486	1.032	0.003
Spending_Game Time Cards	693	2.118	37,486	0.305	0.000
Spending_O2O Store	693	1.665	37,486	1.101	0.269
Spending_Hotel	693	1.052	37,486	0.389	0.009
Spending_Used Good	693	0.606	37,486	0.187	0.040
Spending_Tencent QQ	693	0.467	37,486	0.197	0.387
Spending_Mobile Phone Plans	693	0.167	37,486	0.099	0.229
Spending_Video Game	693	0.137	37,486	0.067	0.632
Spending_Insurance	693	0.096	37,486	0.040	0.101
Spending_Coffee & Beverages	693	0.009	37,486	0.015	0.629
Browsing_SmartDevices	693	0.324	37,486	0.076	0.000
		Panel B: After matching			
Age	693	31.00	3,192	31.07	0.816
Gender (male = 1)	693	0.390	3,192	0.408	0.382
# Historical Purchases	693	560.0	3,192	556.2	0.899
Spending_Mobile Refill Card	693	17.00	3,192	16.88	0.898
Spending_All others	693	3.722	3,192	4.503	0.579
Spending_Transportation	693	2.520	3,192	2.992	0.646
Spending_Movies & Sports Ticket	693	2.348	3,192	2.342	0.988
Spending_Fresh Hema	693	2.123	3,192	1.858	0.630
Spending_Game Time Cards	693	2.118	3,192	0.942	0.214
Spending_O2O Store	693	1.665	3,192	1.974	0.663
Spending_Hotel	693	1.052	3,192	1.298	0.717
Spending_Used Good	693	0.606	3,192	0.318	0.244
Spending_Tencent QQ	693	0.467	3,192	0.165	0.059
Spending_Mobile Phone Plans	693	0.167	3,192	0.249	0.368
Spending_Video Game	693	0.137	3,192	0.164	0.869
Spending_Insurance	693	0.096	3,192	0.076	0.629
Spending_Coffee & Beverages	693	0.009	3,192	0.009	1.000
Browsing_SmartDevices	693	0.324	3,192	0.224	0.224

Note. Calculated over the preadoption period.

propensity score model is independent of the unobserved ε_{it} in the DiD regression model.

Table 3 reports the estimated main effect of consumers' Genie adoption on their weekly spending amount. The results indicate that Genie adoption leads the average adopter to spend RMB 5.719 more per week. Correspondingly, within the first four months after adoption, Genie led to an average increase of 16.6% in weekly spending for adopters in our sample compared with the pretreatment mean of RMB 34.35. The finding of a significant positive effect of Genie adoption on Tmall spending implies that having Genie increases the consumer's tendency to place orders. These results confirm our Hypothesis 1 that voice AI adoption has a positive overall effect on consumer spending.

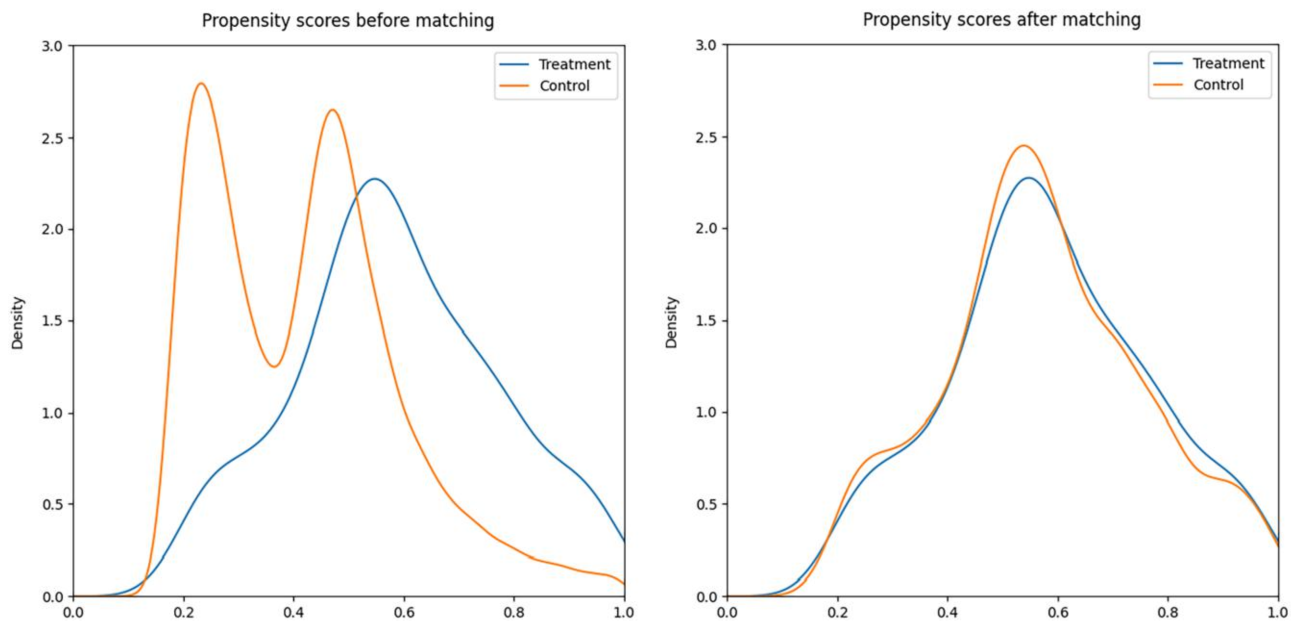
To delve deeper into what drives the increase in spending, we also estimate the impact of Genie adoption on spending in the PC and mobile channels. As shown in columns (2) and (3) in Table 3, Genie adoption has an insignificant (spillover) impact on spending in the mobile channel and a significantly positive (spillover) impact on spending in the PC channel. The residual can thus be attributed to the (direct) effect on the voice channel. Section 7 delves deeper into the effects of channel dynamics.

5. Moderating Effect of Product Category Features

This section delves into the moderating role of product category features on the effect of Genie adoption on spending. These analyses have both practical (for the strategic placement of Genie promotions) and theoretical significance (by offering insight into the

Downloaded from informs.org by [137.68.121.64] on 29 October 2025, at 00:32 . For personal use only, all rights reserved.

Figure 1. (Color online) Distribution of Propensity Scores Before and After Matching



Notes. Logit model with standard errors in parentheses. Estimates are calculated on an unmatched sample of 693 adopters and 37,486 nonadopters in the pretreatment period; the unit of analysis is the individual user. The dependent variable is whether the user adopted Tmall Genie (adoption = 1) during the first week of July 2019 or did not adopt Genie at any time before the end of the sample period (adoption = 0).

underlying mechanism). Specifically, we investigate three potential moderators at the product category level: new versus repeat purchases, product substitutability, and familiarity.

The choices of moderators are informed by the finding that information presented by voice can be more difficult to process than the same information presented in writing (Munz 2020), leading to a higher cost of information acquisition in the voice shopping channel. Products that are familiar, less substitutable, and purchased repeatedly do not require extensive searching or comparison, which makes the shopping process less costly (Adamopoulos et al. 2020, Park et al. 2020). We thus anticipate Genie to be better suited for purchasing these types of products.

To define new versus repeat purchases, we set a prestudy window of six weeks and identify the set of categories from which the consumer has made purchases (denoted as set A).[20] For the preadoption period, we define repeat purchases as those in set A and new purchases as those out of set A (denoted as set B). For the postadoption period, we then define repeat purchases as those in either set A or set B and new purchases as those out of both set A and set B (denoted as set C).[21] We expect the cost associated with ordering repeat purchases will be lower than that for new products, as it costs consumers less time in searching or comparing. We reestimate Equation (2), with the spending on new and repeat purchases as two separate dependent variables.

In Table 4, we find a significantly positive effect on repeat purchases, whereas the effect on new purchases is much smaller for both the overall spending and the voice channel spending, indicating that the positive treatment effect is primarily driven by consumers'

Table 3. Main Effect

Variables	*All Spending* (RMB)	*Mobile Spending* (RMB)	*PC Spending* (RMB)
Adoption × *Post*	5.719**	0.795	2.157***
	(2.018)	(2.510)	(0.690)
Constant	32.92***	27.94***	4.902***
	(0.435)	(0.260)	(0.071)
Individual FEs	Yes	Yes	Yes
Week FEs	Yes	Yes	Yes
Observations	120,435	120,435	120,435
Adjusted R^2	0.138	0.103	0.058

Notes. Robust standard errors are in parentheses. FE, fixed effects.
$*p < 0.05$; $**p < 0.01$; $***p < 0.001$.

Downloaded from informs.org by [137.68.121.64] on 29 October 2025, at 00:32 . For personal use only, all rights reserved.

Table 4. Moderating Role of New vs. Repeat Purchases on Main Effect Across All Channels

	All Channels (RMB)		*Voice Channel* (RMB)[a]	
Variables	*New Purchase*	*Repeat Purchase*	*New Purchase*	*Repeat Purchase*
Adoption × *Post*	0.119	5.600***	0.412***	2.783***
	(1.364)	(1.492)	(0.048)	(0.167)
Constant	12.24***	20.68***	0.000	0.000
	(0.294)	(0.321)	(0.028)	(0.070)
Individual FEs	Yes	Yes	Yes	Yes
Week FEs	Yes	Yes	No	No
Observations	120,435	120,435	21,483	21,483
Adjusted R^2	0.036	0.118	0.022	0.031

Note. Robust standard errors are in parentheses.

[a]The spending in the voice channel for the control group cannot be observed. Therefore, we only keep the observations for the treatment group, and the estimates capture the average spending in the voice channel for the treated users. For other tables involving spending in the voice channel, a similar model specification ($y_{it} = \gamma \cdot Post_t + \delta_i + \varepsilon_{it}$) is applied. We thank an anonymous editor for making this point.

*$p < 0.05$; **$p < 0.01$; ***$p < 0.001$.

repeat purchases, supporting our Hypothesis 2 that the effect of voice AI adoption on spending through the voice channel is stronger for products that do not bear the high cost of information acquisition.[22]

To further test Hypothesis 2, we leverage two additional moderators, product substitutability and familiarity, using data at the user-week-category level.

Substitutability describes the extent to which consumers perceive products as interchangeable with each other. If a product category has *low substitutability*, then each product in the category can be treated as an independent market. Products with low substitutability should not require much active searching or comparison and can therefore be more compatible with voice shopping. We anticipate that the positive effect of Genie adoption on spending is stronger for categories with low substitutability.

We determine which product categories with *low substitutability* by reviewing existing findings. Prior literature has found that books (Chevalier and Goolsbee 2003, McMillan 2007), music (Shiller and Waldfogel 2011), antiques, and artwork (McAfee and McMillan 1987) have sufficiently low substitutability that each product can be treated as an independent market. Additionally, the prescription drug category is included, given that consumers are limited to choosing the product prescribed to them. We estimate the moderating effect of low substitutability with the following equation:

$$y_{ijt} = \gamma \cdot Post_t + \lambda_{LS} \cdot (Post_t \times LowSubstitutability_j) + \delta_i + \sigma_j + \varepsilon_{ijt}, \quad (3)$$

where y_{ijt} is the amount spent by consumer i in week t on category j; σ_j is the category fixed effects; $LowSubstitutability_j$ is a dummy variable that equals one if j belongs to a category with low substitutability, and zero otherwise; γ captures the main effect; and λ_{LS} represents the interaction effect of Genie adoption and low substitutability.

Familiarity, like *low substitutability*, should enable consumers to make purchases without an active search or comparison because consumers would already possess prior knowledge about the product options in the category. We proxy *familiarity* with the number of times a product category j was purchased by consumer i in the pretreatment period.[23] We anticipate that the familiarity of purchases in a product category positively moderates the effect of Genie adoption on spending in that category, and estimate the moderating effect with the following equation:

$$y_{ijt} = \gamma \cdot Post_t + \lambda_F \cdot (Post_t \times Familiarity_{ij}) + \delta_i + \sigma_j + \varepsilon_{ijt}, \quad (4)$$

where y_{ijt} is the amount spent by consumer i in week t on category j, $Familiarity_{ij}$ is the number of times a product category j was purchased by consumer i in the pretreatment period as defined above, γ captures the main effect, and λ_F represents the interaction effect of Genie adoption and familiarity.

To test how substitutability and familiarity of a product category moderate the effect of Genie adoption on spending in the focal category, we estimate Equations (3) and (4) on the purchase data in the voice channel and present the findings in the first two columns of Table 5. Moreover, we also include both moderators in the same model and report the result in the third column of Table 5.[24]

The estimated coefficients of the interaction terms *Post* × *Low Substitutability* and *Post* × *Familiarity* are both positive and significant, supporting our expectation that the impact of Genie adoption on spending through the voice channel is stronger in product categories that are less substitutable and more familiar. These results corroborate with the findings in Table 4 to jointly confirm Hypothesis 2.

Downloaded from informs.org by [137.68.121.64] on 29 October 2025, at 00:32 . For personal use only, all rights reserved.

Table 5. Heterogenous Effect Across Product Category Features

Variables	*Voice Channel* *Spending* (RMB)		
Post	0.000	0.015***	−0.002
	(0.001)	(0.001)	(0.001)
Post × *Low Substitutability*	0.823***		0.654***
	(0.008)		(0.008)
Post × *Familiarity*		0.259***	0.202***
		(0.001)	(0.001)
Constant	0.000	0.000	0.000
	(0.001)	(0.001)	(0.001)
Individual FEs	Yes	Yes	Yes
Category FEs	Yes	Yes	Yes
Observations	2,943,171	2,943,171	2,943,171
Adjusted R^2	0.026	0.026	0.028

Note. Robust standard errors are in parentheses.
$^*p < 0.05$; $^{**}p < 0.01$; $^{***}p < 0.001$.

6. Novelty Effect

Does the positive effect of Genie adoption on spending remain stable over time? To investigate the temporal dynamics of Genie adoption, we estimate the treatment effect in three posttreatment periods: short term (posttreatment week 1, λ_{ST}), medium term (posttreatment weeks 2–7, λ_{MT}), and long term (posttreatment weeks 8–17, λ_{LT}). Following Datta et al. (2018) and Zhou et al. (2020), the three effects on the overall spending and spending in the voice channel are captured using the following equation:

$$\begin{aligned} y_{it} = {} & \lambda_{ST} \cdot (Adoption_i \times Post_t \times I[0 \leq week\_since\_adoption_t \leq 1]) \\ & + \lambda_{MT} \cdot (Adoption_i \times Post_t \times I[2 \leq week\_since\_adoption_t \leq 7]) \\ & + \lambda_{LT} \cdot (Adoption_i \times Post_t \times I[week\_since\_adoption_t \geq 8]) \\ & + \delta_i + \tau_t + \varepsilon_{it}, \end{aligned} \tag{5}$$

where y_{it} is the total amount spent by consumer i in week t, and $I[week\_since\_adoption_t]$ is the indicator variable. In Table 6, we report the estimated treatment effects in the short, medium, and long term on the overall spending in column (1). We also reestimate Equation (5) to observe the changes in treatment effect over time while taking into consideration the moderating effect of product features, with the subsets of new (column (2)) and repeat purchases (column (3)) in the voice shopping channel.

We find that the overall treatment effect is the strongest in the short term (+RMB 10.17 weekly per user). The effect size diminishes in the medium term (+RMB 7.796 weekly per user) and becomes insignificant in the long term. Breaking down the overall spending, we learn from column (3) that the treatment effects are positive and significant throughout the posttreatment period on repeat purchase in the voice channel. The treatment effect is the strongest in the short term (+RMB 8.676 weekly per user in the short term),[25] diminishes in the medium term (+RMB 2.893 weekly per user), and diminishes further in the long term (+RMB 1.538 weekly per user). It is likely that the novelty effect is at work—consumers who adopt Genie are initially enticed by a curiosity about the novel AI gadget and/or by an urge to get familiar with the product. As the product becomes more familiar, consumers settle into a more modest pattern of increased spending and may eventually revert to their purchase habits before Genie adoption. Moreover, there is a positive and significant impact on new purchase (column (2)) in the short term, though the effects are comparatively smaller than those on repeat purchase. The novelty of voice shopping technology to the consumers can encourage them to explore a wider range of new products, but such an effect is small and decays over time,

Table 6. Novelty Effect

Variables	(1) All *Total Spending* (RMB)		(2) Voice[a] *New Purchase* (RMB)	(3) Voice *Repeat Purchase* (RMB)
Adoption × *Post* × *Short term*	10.17*	*Post* × *Short term*	2.263***	8.676***
	(4.211)		(0.355)	(0.599)
Adoption × *Post* × *Medium term*	7.796**	*Post* × *Medium term*	0.252***	2.893***
	(2.736)		(0.068)	(0.219)
Adoption × *Post* × *Long term*	3.583	*Post* × *Long term*	0.137***	1.538***
	(2.332)		(0.037)	(0.207)
Constant	32.92***	*Constant*	0.000	0.000
	(0.435)		(0.028)	(0.097)
Individual FEs	Yes	Individual FEs	Yes	Yes
Week FEs	Yes	Week FEs	No	No
Observations	120,435	Observations	21,483	21,483
Adjusted R^2	0.138	Adjusted R^2	0.013	0.052

Note. Robust standard errors are in parentheses.
[a]The model specification for columns (2) and (3) is the same as Equation (5) except that *Adoption* = 1 and week FEs are not included.
$^*p < 0.05$; $^{**}p < 0.01$; $^{***}p < 0.001$.

Downloaded from informs.org by [137.68.121.64] on 29 October 2025, at 00:32 . For personal use only, all rights reserved.

as shown by the even smaller coefficients in the medium and long term.

Overall, our findings reveal that the voice assistant device promotes increased repeat purchases in the short run but fails to retain the same level of interest over the long run, aligning with the concept of a novelty effect. By comparing the magnitude of the effects on the total spending and the spending on the voice channel, we find that in the short term, the sum of the effects on new purchases and repeat purchases in the voice channel is close to the effect on the total spending, indicating that the boost in overall spending in the short term is mainly attributed to the increased spending in the voice channel. However, in the medium and long terms, the sum of the effects on new purchases and repeat purchases in the voice channel cannot fully account for the effect on the total spending, suggesting that there could be a cross-channel spillover effect because of Genie adoption, which we further investigate in the next section.

7. Channel Dynamics

The findings presented so far indicate a positive effect of Genie adoption on Tmall spending. By providing a new channel for shopping, Genie has the potential to cause spillover effects on spending in the PC and mobile channels. Table 3 further shows that the magnitude of the spillover effect is roughly comparable to that of the direct effect. This section proceeds to examine if the adoption leads to an increase in spending across channels (i.e., a complementary effect) or if it merely prompts consumers to switch their spending from the PC and mobile channels to the voice channel (i.e., a cannibalization effect). To test Hypothesis 4, we start by reestimating Equation (2), the main effect model, using the purchase data from the mobile and PC channels, as shown in Table 7. Additionally, we further divide the purchases into new and repeat purchases to get a more comprehensive view of the estimated treatment effect for each channel.

From Table 7, we can derive the following conclusions regarding the overall spillover effects onto the other two channels. First, there is no significant negative treatment effect, so the voice channel does not seem to be cannibalizing the other two channels. Instead, in the PC channel, we observe positive and significant effects on spending on repeat purchases (column (4)) in the medium (+RMB 2.155, $p < 0.01$) and long term (+RMB 1.312, $p < 0.05$), whereas the effect on new spending in the PC channel is insignificant (column (3)). Genie adoption has a significant positive spillover effect on repeat purchases in the PC channel starting from the medium term. We reason that as a consumer develops more trust and familiarity with the voice AI over time, one may engage in more conversations related to personal life and interests.[26] In the case where the adopters want to proceed with the purchase process after the conversation with voice AI but are constrained by the high cost of searching and comparing products using voice, they would direct themselves to other shopping channels to complete the purchase process. Particularly for the PC channel, because of its larger screen size and more thorough visual representation of product information, the positive spillover effect is mainly oriented here. Last but not least, the overall spillover effect of Genie adoption on spending in the mobile channel is not significant, regardless of new or repeat purchases. The insignificant effect could be attributed to a recognized limitation of the mobile channel: the small screen size displays fewer product

Table 7. Heterogeneity in the Spillover Effect Across Channels

	Mobile		PC	
Variables	(1) *New Purchase* (RMB)	(2) *Repeat Purchase* (RMB)	(3) *New Purchase* (RMB)	(4) *Repeat Purchase* (RMB)
Adoption × Post × Short term	−0.906	−1.108	0.819	0.850
	(2.586)	(2.859)	(1.148)	(1.124)
Adoption × Post × Medium term	0.604	1.645	0.679	2.155**
	(1.680)	(1.857)	(0.746)	(0.730)
Adoption × Post × Long term	−1.336	1.821	0.537	1.312*
	(1.432)	(1.583)	(0.636)	(0.622)
Constant	9.810***	18.13***	2.381***	2.521***
	(0.267)	(0.295)	(0.119)	(0.116)
Individual FEs	Yes	Yes	Yes	Yes
Week FEs	Yes	Yes	Yes	Yes
Observations	120,435	120,435	120,435	120,435
Adjusted R^2	0.032	0.114	0.010	0.062

Note: Robust standard errors are in parentheses.
*$p < 0.05$; **$p < 0.01$; ***$p < 0.001$.

Downloaded from informs.org by [137.68.121.64] on 29 October 2025, at 00:32 . For personal use only, all rights reserved.

options than a PC, which makes the search harder (Ghose et al. 2013).

To attain a better comprehension of the underlying mechanisms engendering spillover effects, we assess treatment effects at the category level within each channel (see results in Online Appendix A12). Although statistically significant effects are absent in most categories, Genie adoption manifests significant effects in four specific categories, consistent with the argument posited in Hypothesis 4. Notably, within the PC channel, the adoption of Genie leads to a significant reduction in expenditure on *Mobile Refill Cards* and a significant increase in expenditure on *Movie & Sports Tickets*. This attests to the cannibalization effect of Genie adoption on the acquisition of familiar items, which reduces transaction costs for consumers (Park et al. 2020). Conversely, concerning unfamiliar items, Genie adoption exerts a complementary effect on the PC channel, where consumers benefit from an enhanced transactional experience facilitated by the expanded screen (Ghose et al. 2013).[27] Within the mobile channel, a negative spillover effect is found in the spending on *Online Game Point Cards*, a transaction conveniently executed hands-freely through the voice channel (Bahmani et al. 2022). By contrast, there is a positive spillover effect on the spending on *Snacks, Nuts & Specialties*. This divergence emerges as consumers, after initiating dialogues with voice AI, transition to the mobile channel to complete the purchases, motivated by the high costs of product comparisons via voice (Xu et al. 2017).[28] Taken together, these findings provide empirical evidence for the channel dynamics proposed in Hypothesis 4.

8. Post Hoc Analyses

To derive further insights into the potential impacts of Genie adoption, we conduct a series of post hoc analyses to examine: (1) the heterogeneity across user demographics, (2) channel dynamics for middle-aged consumers, and (3) the treatment effect on price and purchase quantity, which may have driven the effect on spending. Additionally, we conduct several robustness checks.

8.1. Heterogeneity in the Treatment Effect Across User Demographics

The impact of Genie adoption on spending potentially varies depending on consumer characteristics, notably age and gender. It is also interesting and socially meaningful to investigate and provide more theoretical substantiation for these variations, as it could provide a deeper understanding of how different segments of consumers interact with this technology and how it affects their spending patterns. We investigate the heterogeneity in the treatment effect across consumer age and gender in Online Appendix A5.

8.2. Middle-Aged Consumers' Channel Dynamics

As presented in Online Appendix A5, the effect of Genie adoption on purchase spending is negative for consumers older than 45 years of age.[29] As technology continues to advance, it can be challenging for older individuals to keep up and take advantage of the latest developments. From a social equity perspective, it is important to investigate the changes behind the overall decrease in spending for this group of adopters. This can complement the factors that researchers and developers should take into consideration when designing inclusive AI systems, such as the need for accessibility, ease of use, and the ability to interact with technology in a natural and intuitive way. We select the consumers of age ≥45 in the treatment group together with their matched consumers in propensity score matching. We present the findings in Online Appendix A6.

8.3. Total Quantity and Average Price

To more thoroughly understand the mechanism driving increased spending, we expand our main analysis to include two additional dependent variables: the total quantity and the average price of items purchased by users. This approach will enable us to determine whether the increased spending is driven by consumers buying more items, at a higher price point, or a combination of both factors.

We first examine the impact of Genie adoption on the average price of products purchased across shopping channels over time. In Table 8, the estimated treatment effects on the average price of products purchased in mobile and PC channels are insignificant in general (columns (1) to (4)). This indicates that Genie adoption does not lead consumers to purchase products of higher prices.

We then examine the treatment effects on total purchase quantity across channels over time. In Table 9, we observe positive and significant effects on the quantity of repeat purchases in the PC channel, and the effects persist across different time periods (column (4)). For the purchase quantity of new products, we observe a marginally significant quantity increase in the mobile channel. Together with the results reported in Tables 5 and 7, our findings imply that the positive spillover effects in the repeat purchase via the PC channel are driven by the increasing purchase quantity rather than a higher price. Additionally, the overall positive impact on spending across all channels is largely attributed to consumers buying repeatedly purchased products with higher prices through the voice channel.

There might exist an alternative explanation that voice AI may recommend users deals or promotions with higher prices, thereby driving increased spending. However, this possibility can be ruled out, as the Genie voice AI recommends the same set of products

Downloaded from informs.org by [137.68.121.64] on 29 October 2025, at 00:32 . For personal use only, all rights reserved.

Table 8. Treatment Effect on Average Price

	Average Price (RMB)					
	Mobile		PC		All	
Variables	(1) *New Purchase*	(2) *Repeat Purchase*	(3) *New Purchase*	(4) *Repeat Purchase*	(5) *New Purchase*	(6) *Repeat Purchase*
Adoption × *Post* × *Short-term*	−0.070	−1.267	0.830	0.150	1.807	1.869
	(1.473)	(1.611)	(0.766)	(0.861)	(1.584)	(1.706)
Adoption × *Post* × *Medium-term*	0.959	0.771	0.593	0.589	1.479	2.239*
	(0.957)	(1.047)	(0.497)	(0.559)	(1.029)	(1.109)
Adoption × *Post* × *Long-term*	−0.524	1.269	0.717	0.362	0.176	2.010*
	(0.816)	(0.892)	(0.424)	(0.477)	(0.877)	(0.945)
Constant	4.301***	13.04***	1.649***	2.194***	5.503***	14.481***
	(0.152)	(0.166)	(0.079)	(0.089)	(0.164)	(0.176)
Individual FEs	Yes	Yes	Yes	Yes	Yes	Yes
Week FEs	Yes	Yes	Yes	Yes	Yes	Yes
Observations	120,435	120,435	120,435	120,435	120,435	120,435
Adjusted R^2	0.007	0.120	0.002	0.055	0.009	0.121

Notes. Results for the voice channel are omitted as the average pretreatment item price for adopters and nonadopters could not be observed. Robust standard errors are in parentheses.
$*p < 0.05$; $**p < 0.01$; $***p < 0.001$.

regardless of the channel in which consumers conduct their search (we present further details in Online Appendix A2). The above analyses on the total quantity and average price of items purchased further help rule out this alternative explanation. Besides, Tmall Genie will not initiate any reminders if users do not enter any search queries. Instead, users use Tmall Genie to buy items that they are familiar with. Therefore, the increased spending is not a result of any reminder effect.

8.4. Robustness Checks

To further validate our main results, we conducted a series of robustness checks, falsification tests, and additional empirical analyses.

8.4.1. Placebo Treatments Test. As the DiD approach requires the assumption of parallel pretreatment trends, we validate this assumption by conducting a placebo treatment test. Specifically, we define a set of placebo "treatments" at three midpoints in the pretreatment period (namely, weeks 6, 7, and 8), and we reestimate the DiD model on the same matched sample prior to the actual treatment week. For example, if we set week 6 as the placebo treatment week, then we use weeks 1–5 as the pretreatment period and weeks 7–13 as the posttreatment period. Theoretically, the effect of these placebo "treatments" should be insignificant, given that they are not actual interventions and no real treatment occurred during this period (i.e., before week

Table 9. Treatment Effect on Total Quantity

	Total Quantity					
	Mobile		PC		All	
Variables	(1) *New Purchase* (RMB)	(2) *Repeat Purchase* (RMB)	(3) *New Purchase* (RMB)	(4) *Repeat Purchase* (RMB)	(5) *New Purchase* (RMB)	(6) *Repeat Purchase* (RMB)
Adoption × *Post* × *Short-term*	0.106	1.669	0.009	0.045***	0.173	1.984
	(0.255)	(1.082)	(0.006)	(0.009)	(0.255)	(1.082)
Adoption × *Post* × *Medium-term*	0.356*	0.348	0.002	0.020***	0.362*	0.431
	(0.166)	(0.703)	(0.004)	(0.006)	(0.166)	(0.703)
Adoption × *Post* × *Long-term*	0.118	0.336	0.001	0.014**	0.120	0.378
	(0.141)	(0.599)	(0.003)	(0.005)	(0.141)	(0.599)
Constant	0.131***	1.545***	0.023***	0.043***	0.154***	1.589***
	(0.026)	(0.112)	(0.001)	(0.001)	(0.026)	(0.112)
Individual FEs	Yes	Yes	Yes	Yes	Yes	Yes
Week FEs	Yes	Yes	Yes	Yes	Yes	Yes
Observations	120,435	120,435	120,435	120,435	120,435	120,435
Adjusted R^2	0.001	0.147	0.011	0.202	0.001	0.148

Note. Robust standard errors are in parentheses.
$*p < 0.05$; $**p < 0.01$; $***p < 0.001$.

Downloaded from informs.org by [137.68.121.64] on 29 October 2025, at 00:32 . For personal use only, all rights reserved.

13). The empirical evidence fails to reject the null hypothesis of no treatment effect for the placebo "treatments" (see results in Online Appendix A7).

8.4.2. Temporal Selection Test and Dynamic Effects. There may exist a within-consumer temporal selection issue that affects the timing of Genie adoption. If time-varying unobserved factors systematically prompt consumers to buy Genie concomitantly with other products on Tmall, then the increase in spending after adoption might not be attributable to Genie adoption. For instance, if a consumer relocates to a new house, the move may prompt them to buy a Genie. However, the consumer may also need to buy many other products because of the move. In this case, the true but unobserved cause of their increased spending would be the move, rather than the adoption of the Genie. We address this concern by incorporating an additional interaction term into the DiD model:

$$y_{it} = \beta \cdot Post_t + \gamma \cdot (Adoption_i \times Post_t) + \rho \cdot (Adoption_i \times 1week\_pre_t) + \delta_i + \tau_t + \varepsilon_{it}, \quad (6)$$

where $1week\_pre_t$ is an indicator of the week prior to the adoption week. If an unobserved factor is driving both adoption and increased spending, then $Adoption_i \times 1week\_pre_t$ should be positive. However, the result (reported in Online Appendix A8) shows a negative and insignificant coefficient of $Adoption_i \times 1week\_pre_t$. We conclude that within-consumer temporal selection is not a major threat to our identification.

Furthermore, we estimate a relative time model (Autor 2003, Greenwood and Wattal 2017), where the event is defined as the adoption of the voice AI (i.e., the 14th week).[30] As a standard procedure, the first lag (one week before the event) is set as the baseline and therefore omitted. We then estimate the following model:

$$\begin{aligned} y_{it} = {} & \lambda_{LT}^{Pre} \cdot Adoption_i \times Pre_t \times LongTerm_t \\ & + \lambda_{MT}^{Pre} \cdot Adoption_i \times Pre_t \times MediumTerm_t \\ & + \lambda_{ST}^{Pre} \cdot Adoption_i \times Pre_t \times ShortTerm_t \\ & + \lambda_{ST}^{Post} \cdot Adoption_i \times Post_t \times ShortTerm_t \\ & + \lambda_{MT}^{Post} \cdot Adoption_i \times Post_t \times MediumTerm_t \\ & + \lambda_{LT}^{Post} \cdot Adoption_i \times Post_t \times LongTerm_t \\ & + \delta_i + \tau_t + \varepsilon_{it}. \end{aligned} \quad (7)$$

The estimation results in Table A8.2 in the Online Appendix present the dynamic effects over time. All the coefficients of the pretreatment dummies are statistically indistinguishable from zero, confirming the absence of pretrends. The magnitude and statistical significance of the posttreatment dummies decline over time, which indicates the diminishing positive effect of the Genie adoption.

8.4.3. An Alternative Aggregation Level of the Category. For the main estimation, we calculated propensity scores based on purchases in the top 13 categories plus a 14th "*all others*" category (aggregated from the remaining 124 categories). Though we believe that the granularity of this calculation is effective for identifying similar consumers, we conduct an alternative specification with less granularity to ensure the robustness of the findings. Specifically, we aggregate all products into just four categories: electronics, travel services, food, and all others. We repeat the matching procedure and DiD estimation. The results (reported in Online Appendix A9) are qualitatively the same as the main results.

8.4.4. An Alternative Measure of New vs. Repeat Purchases. Additionally, we use an alternative measure of new versus repeat purchases. Specifically, we calculate the proportional changes defined with the average spending amount of an individual on new and repeat purchases before the adoption as the baseline.[31] We then reestimate the model and report the result in Online Appendix A10. The analysis reveals that the treatment effect on new purchases is not statistically different from zero, whereas the treatment effect on repeat purchases is significantly positive, thereby confirming our main results.

8.4.5. Different Prestudy Windows in Defining New vs. Repeat Purchases. In previous sections, we define new versus repeat purchases using a prestudy window of six weeks. To ensure the robustness of the selection of time window, we also reestimate all the models relying on such a definition using prestudy windows of three, four, five, seven, and eight weeks. The results are all qualitatively similar, as shown in Online Appendix A11.

8.4.6. An Additional Data Set and the IV Method. As described in Section 3.1, a potential concern with our empirical setting is that consumers may choose to adopt Genie because of some nonrandom shocks that are unobservable (and thus cannot be incorporated into our matching procedure) and also cannot be absorbed by the individual and time fixed effects. In Online Appendix A13, we provide extra empirical evidence using an additional data set collected from Alibaba in a different time period; we take advantage of an essentially random coupon allocation event to use an instrumental variable approach. The estimated main effect is qualitatively consistent with the effect presented in the main text.

Downloaded from informs.org by [137.68.121.64] on 29 October 2025, at 00:32 . For personal use only, all rights reserved.

9. Implications and Conclusions

Voice shopping holds increasing significance in our daily lives and the digital economy. This paper takes the first step toward understanding the impact of consumer adoption of voice shopping technology on subsequent spending within the affiliated e-commerce platform. We employ granular propensity score matching to match adopters with nonadopters and thereafter use the difference-in-differences approach to identify the effect of Genie adoption on the e-commerce spending.

Our analyses yield the following main findings: first, our findings indicate that the adoption of Tmall Genie, the voice AI device launched by e-commerce giant Alibaba, leads consumers in our sample to increase their average weekly spending on Tmall by 16.6% within the first four months after adoption. Second, Genie adoption has stronger positive effects on spending for repeat purchases and less substitutable and more familiar products, all of which do not require consumers to digest product information at high costs. Constrained by the inconvenience of searching and comparing products using voice, consumers prefer to purchase products that do not require complicated selection processes through the voice channel. Third, whereas the positive effect of Genie adoption on spending attenuates over time, it continues to be significantly positive in the long term, especially for repeat purchases in the voice channel. Because of the novelty effect and the shortcomings of information processing using voice, the long-term positive effect is mainly attributed to repeat purchases. Lastly, Genie adoption produces a positive and significant spillover effect on spending via the PC channel, whereas its impact on mobile channel spending is insignificant. The category-level analysis illustrates that the direction of the spillover effect on the two channels, whether positive or negative, depends on the specific shopping context. Overall, the impact of voice AI on purchases can be broken down into direct and spillover impacts, and the magnitude of the spillover effect is comparable to that of the direct effect.

Our study provides several managerial implications for the development of voice AI technology in e-commerce platforms. First, we demonstrate that the adoption of a voice AI device facilitates incremental online spending, so voice AI technologies may be well positioned to enhance the revenue growth of e-commerce platforms. When designed well, voice AI devices, as a new form of consumer-platform interaction, can improve user experience and help companies build winning relationships with their customers,[32] but there can be challenges in terms of information processing that limit its potential.

Our findings show that the positive effects of voice AI adoption on spending are mainly attributed to the product categories that do not require consumers to do extensive search or comparison, namely, repeat purchases and more familiar and less substitutable product categories. To realize the fuller potential of voice AI technology, future product developers could take this and be cognizant of the cost of information processing in their product design. For instance, future designs could include features such as small screen displays on the Genie devices to complement the shortcomings of voice AI, such as unclear or unnatural representation of information. This can help decrease consumers' perceived amount of effort needed to interact with voice AI, and therefore, they could adopt voice shopping for a wider range of products. Another way is to present choices and highlight information in a manner that consumers can easily understand and evaluate through speech. For example, companies can leverage the purchase histories of Genie users and personalize product recommendations for repeat purchases and more familiar products via the voice marketing channel. Such targeted marketing practices could potentially lead to reduced marketing expenditures and a higher return on investment for managers.

Second, our results show that the adoption of voice shopping has a positive spillover effect on spending in the PC channel (and no significant effect on spending in the mobile channel). One possible cause of the complementary relationship is that consumers may combine the voice channel with another channel to complete a purchase. Managers may want to keep this positive spillover effect in mind when designing e-commerce platforms for optimized customer engagement (Sun 2025). Moreover, our study finds that engagement and motivation for continued usage of voice AI decline over time because of the novelty effect. Future product designs can focus on creating more consumer-oriented technology, which can hopefully maintain consumers' interest in using voice AI. For instance, companies can adjust the emotion-expressing capabilities (Han et al. 2022) and incorporate human-like behavior into AI agents (Schanke et al. 2021) to improve their performance and efficacy so that even after the short-term curiosity disappears, the higher perceived usefulness and better human-computer interaction can remain attractive to consumers in the long run. Lastly, our research extends its implications beyond the realm of e-commerce to other business sectors. For example, in the hospitality industry, Marriott hotels now include Amazon Alexa; guests can order room service by talking to a voice-activated ordering device in their rooms. The heterogeneity and moderating effects we document in the effects of voice-activated shopping devices may transfer to offline voice-aided purchases (or demand for services) in the context of hotels.

We conclude by identifying several questions beyond this paper's scope, yet they represent potential avenues for future research. First, we are unable to access the specific product information, but only category information, for each transaction. More granular data on consumers' purchase journeys and transaction

Downloaded from informs.org by [137.68.121.64] on 29 October 2025, at 00:32 . For personal use only, all rights reserved.

details, particularly the search aspect, would have allowed for a deeper exploration of the heterogeneous effects and the nuanced mechanisms between channels. Second, as consumers become more familiar with AI products, the impact of voice AI device adoption on shopping behaviors may diminish. Our findings on the short-term and long-term effects partially address this issue, but we urge caution in extrapolating the effect to an even longer term (i.e., beyond 17 weeks postadoption). Third, though we have leveraged several econometric techniques to address the potential endogeneity concerns, more can be done to more accurately identify the impact of Genie adoption on a consumer's Tmall spending, such as analyses on adopters who received the device without self-selection. Moreover, to provide more evidence on the underlying mechanism behind the channel spillover effects, future researchers could exploit more granular data on consumers' cross-channel purchase journeys, which we are unfortunately unable to collect and process. Lastly, in this study, we are agnostic about how voice AI technology can better serve consumers, but we hope future research can explore this question with richer data of full conversations between voice AI agents and consumers.

Acknowledgments

The authors gratefully acknowledge Feiyu Xiong, Xueying Li, and Ying Bao from the Alibaba Group for providing the access to the raw data set utilized in this study. The authors thank Yu Zhang and Yuxin Geng for their research assistance.

Endnotes

[1] Major features and uses of four popular voice assistant devices are provided in Online Appendix A1.

[2] See https://capitaloneshopping.com/research/voice-shopping-statistics/.

[3] See https://www.racked.com/video/2018/2/20/17031958/alexa-shopping-amazon-echo-dot-voice.

[4] See https://www.walmart.com/cp/voice-shopping/5028404.

[5] See https://debutify.com/blog/voice-shopping/.

[6] See https://voicebot.ai/2017/08/02/voice-meets-impulse-buy-thought-transaction-22-6-seconds/.

[7] See https://www.alibabagroup.com/en/news/press_pdf/p190515.pdf.

[8] See https://www.iimedia.cn/c1061/71838.html.

[9] We use "Genie" as a shorthand for "Tmall Genie," the focal voice-AI device.

[10] See https://www.alizila.com/millions-in-china-tapped-voice-shopping-during-11-11-tmall-genie/.

[11] See https://economictimes.indiatimes.com/news/international/business/alibaba-to-invest-1-4-billion-in-ai-system-for-smart-speakers/articleshow/75842991.cms?from=mdr.

[12] We provide two additional examples of shopping-related conversations with Genie in Online Appendix A2.

[13] Given the small fraction of voice-assisted purchases in our data set, our estimated effects may be most applicable to the phenomenon of early penetration.

[14] See https://www.sec.gov/Archives/edgar/data/1577552/000110465920082881/a20-6321_46k.pdf.

[15] Our data do not include those whose first-time adoption happened in other weeks. In our data, all adopters adopted Genie only once. We provide model-free evidence on the adoption rate by gender and by age group in Online Appendix A3.

[16] We further discuss how we address the temporal selection concern in Section 8.4.2.

[17] We thank the anonymous review team for raising this concern and providing a way to address it.

[18] Specifically, we needed at least 285 purchase records per category for matching.

[19] As a robustness check, we redo the matching procedure with four less granular categories, and we reestimate the DiD model on the new matched sample. The results are qualitatively the same. See details in Section 8.4.3.

[20] We also set the prestudy window as three, four, five, seven, and eight weeks and obtain consistent results. The corresponding results are reported in the robustness check in Section 8.4.5 and Online Appendix A11.

[21] An alternative way is to define the new purchase as the first item purchased in each product category and the repeat purchase as the subsequent purchases in each category. For instance, if user i in week t purchases a product of category j, which has never been purchased before, the amount spent on this product counts toward the spending on new purchase, whereas when the same user i later purchases any product of category j, the amount spent counts as spending on repeat purchase. However, this could result in selection bias that over time, there are fewer (more) categories from which the purchases would qualify for new (repeat) purchases. We thank an anonymous reviewer for pointing out this problem and suggesting a method to address it.

[22] We acknowledge our data limitation that we don't have access to historical data before the data collection period. The way we define new and repeat purchases is thus subject to this limitation. Given that the entire purchase history is not available, the number of categories in set B and set C could be larger than that using the entire purchase history. Such a data limitation has two implications for new and repeat purchases: (1) the effect of voice AI adoption on new purchases could be overestimated. Nevertheless, we find insignificant treatment effects on new purchases across different prestudy time window specifications. Therefore, the actual effect on new purchases should still be insignificant; and (2) the effect of voice AI adoption on repeat purchases could be underestimated. That is, the actual effect on repeat purchases should be larger. If the entire purchase history can be provided, the difference between the effects on new purchases and repeat purchases could be greater, further supporting our Hypothesis 2 that the effect of voice AI adoption on spending through the voice channel is stronger for products that do not bear high costs of information acquisition. We thank an anonymous reviewer for suggesting this discussion.

[23] We thank an anonymous reviewer for suggesting this measurement.

[24] We thank an anonymous reviewer for this suggestion. Besides, the correlation coefficient between substitutability and familiarity is 0.3395.

[25] We discovered that the majority of products purchased by consumers using Genie are of fixed demand, such as mobile refill cards. The sharp increase in short-term spending in the voice channel, which subsequently diminishes, may be because of consumers switching from alternative platforms to the focal platform. Unfortunately, without data from multiple online shopping platforms, we are unable to verify this possibility.

[26] Because all purchases start from consumer-initiated conversations and Tmall Genie cannot initiate any conversation on its own,

Downloaded from informs.org by [137.68.121.64] on 29 October 2025, at 00:32 . For personal use only, all rights reserved.

we expect most conversation topics are those that the consumers have prior knowledge or interest in, rather than new topics that can potentially stimulate spending on new purchases and, hence, the insignificant spillover effect on new spending in the PC channel.

[27] In Online Appendix A12, we provide an example of the *Movie & Sports Tickets* category to highlight the potential challenges in relying solely on the voice channel for transactions.

[28] More detailed data on consumers' purchase journeys, particularly the search aspect, would have allowed for a deeper exploration of the complementarity mechanisms between channels. Regrettably, we are unable to obtain such finer-grained data.

[29] The maximum age in our sample is 57. According to the definition from Wikipedia (https://en.wikipedia.org/wiki/Middle_age), those older than 45 but below 60 belong to the "middle age" group.

[30] Our choice to aggregate the data into short-, mid-, and long-term periods was driven by the observation of significant fluctuations in the week-by-week results. These fluctuations could potentially make the results more difficult to interpret. We thank an anonymous reviewer for this suggestion.

[31] We thank an anonymous reviewer for this suggestion.

[32] See press at https://www.forbes.com/sites/forbestechcouncil/2021/08/24/harnessing-conversational-voice-ai-in-the-e-commerce-industry/?sh=432677f060ec.

References

Adamopoulos P, Todri V, Ghose A (2020) Demand effects of the internet-of-things sales channel: Evidence from automating the purchase process. *Inform. Systems Res.* 32(1):238–267.

Autor DH (2003) Outsourcing at will: The contribution of unjust dismissal doctrine to the growth of employment outsourcing. *J. Labor Econom.* 21(1):1–42.

Ba W, Mendelson H, Zhu M (2020) Sales policies for a virtual assistant. Preprint, submitted September 2, https://arxiv.org/abs/2009.03719.

Bahmani N, Bhatnagar A, Gauri D (2022) Hey, Alexa! What attributes of Skills affect firm value? *J. Acad. Marketing Sci.* 50(6): 1219–1235.

Bang Y, Lee DJ, Han K, Hwang M, Ahn JH (2013) Channel capabilities, product characteristics, and the impacts of mobile channel introduction. *J. Management Inform. Systems* 30(2):101–126.

Benabou R (1993) Search market equilibrium, bilateral heterogeneity, and repeat purchases. *J. Econom. Theory* 60(1):140–158.

Bergner AS, Hildebrand C, Häubl G (2023) Machine talk: How verbal embodiment in conversational AI shapes consumer–brand relationships. *J. Consum. Res.* 50(4):742–764.

Bronnenberg B, Dubé JP, Mela C (2010) Do digital video recorders influence sales? *J. Marketing Res.* 47(6):998–1010.

Brynjolfsson E, Li D, Raymond LR (2023) Generative AI at work. NBER Working Paper No. 31161, National Bureau of Economic Research, Cambridge, MA.

Chen G, Xiao S, Zhang C, Zhao H (2023) A theory-driven deep learning method for voice chat–based customer response prediction. *Inform. Systems Res.* 34(4):1513–1532.

Chevalier J, Goolsbee A (2003) Measuring prices and price competition online: Amazon.com and BarnesandNoble.com. *Quant. Marketing Econom.* 1(2):203–222.

Datta H, Knox G, Bronnenberg BJ (2018) Changing their tune: How consumers' adoption of online streaming affects music consumption and discovery. *Marketing Sci.* 37(1):5–21.

Forbes (2023) Reshaping E-commerce: The influence of AI-Generated content. *Forbes* (August 2), https://www.forbes.com/sites/shoplazza/2023/08/02/reshaping-e-commerce-the-influence-of-ai-generated-content/.

Ghose A (2018) *TAP: Unlocking the Mobile Economy* (MIT Press, Cambridge, MA).

Ghose A, Goldfarb A, Han SP (2013) How is the mobile internet different? Search costs and local activities. *Inform. Systems Res.* 24(3):613–631.

Greenwood BN, Wattal S (2017) Show me the way to go home. *Management Inform. Systems Quart.* 41(1):163–188.

Goli A, Singh A (2024) Frontiers: Can large language models capture human preferences? *Marketing Sci.* 43(4):709–722.

Han E, Yin D, Zhang H (2022) Bots with feelings: Should AI agents express positive emotion in customer service? *Inform. Systems Res.* 34(3):1296–1311.

Hu J, Goh KY, Sun C (2024) Does human-AI co-creation drive wage premium? Evidence from freelance marketplaces. Working paper, National University of Singapore, Singapore.

Huang MH (2003) Designing website attributes to induce experiential encounters. *Comput. Human Behav.* 19(4):425–442.

Kleijnen M, De Ruyter K, Wetzels M (2007) An assessment of value creation in mobile service delivery and the moderating role of time consciousness. *J. Retailing* 83(1):33–46.

Kumar V, Dixit A, Javalgi RRG, Dass M (2016) Research framework, strategies, and applications of intelligent agent technologies (IATs) in marketing. *J. Acad. Marketing Sci.* 44(1):24–45.

Li P, Castelo N, Katona Z, Sarvary M (2024) Frontiers: Determining the validity of large language models for automated perceptual analysis. *Marketing Sci.* 43(2):254–266.

Luo X, Qin MS, Fang Z, Qu Z (2021) Artificial intelligence coaches for sales agents: Caveats and solutions. *J. Marketing* 85(2):14–32.

Luo X, Tong S, Fang Z, Qu Z (2019) Frontiers: Machines vs. humans: The impact of artificial intelligence chatbot disclosure on customer purchases. *Marketing Sci.* 38(6):937–947.

Manchanda P, Packard G, Pattabhiramaiah A (2015) Social dollars: The economic impact of customer participation in a firm-sponsored online customer community. *Marketing Sci.* 34(3):367–387.

Massetti B (1996) An empirical examination of the value of creativity support systems on idea generation. *Management Inform. Systems Quart.* 20(1):83–97.

Mayya R, Viswanathan S (2024) Delaying informed consent: An empirical investigation of mobile apps' upgrade decisions. *Management Sci.*, ePub ahead of print December 2, https://doi.org/10.1287/mnsc.2022.00334.

McAfee RP, McMillan J (1987) Auctions and bidding. *J. Econom. Lit.* 25(2):699–738.

McMillan R (2007) *Different Flavor, Same Price: The Puzzle of Uniform Pricing for Differentiated Products* (Federal Trade Commission, Washington, DC).

Melumad S (2023) Vocalizing search: How voice technologies alter consumer search processes and satisfaction. *J. Consum. Res.* 50(3):533–553.

Meuter ML, Ostrom AL, Roundtree RI, Bitner MJ (2000) Self-service technologies: Understanding customer satisfaction with technology-based service encounters. *J. Marketing* 64(3):50–64.

Munz KP (2020) Not-so easy listening: Roots and repercussions of auditory choice difficulty in voice commerce. Doctoral dissertation, New York University, New York.

Pamuru V, Khern-am-nuai W, Kannan K (2021) The impact of an augmented-reality game on local businesses: A study of Pokémon Go on restaurants. *Inform. Systems Res.* 32(3):950–966.

Park Y, Bang Y, Ahn JH (2020) How does the mobile channel reshape the sales distribution in e-commerce? *Inform. Systems Res.* 31(4):1164–1182.

Parthasarathy M, Bhattacherjee A (1998) Understanding post-adoption behavior in the context of online services. *Inform. Systems Res.* 9(4):362–379.

Schanke S, Burtch G, Ray G (2021) Estimating the impact of "humanizing" customer service chatbots. *Inform. Systems Res.* 32(3):736–751.

Shiller B, Waldfogel J (2011) Music for a song: An empirical look at uniform pricing and its alternatives. *J. Indust. Econom.* 59(4):630–660.

Downloaded from informs.org by [137.68.121.64] on 29 October 2025, at 00:32 . For personal use only, all rights reserved.

Smith KT (2020) Marketing via smart speakers: What should Alexa say? *J. Strategic Marketing* 28(4):350–365.

Son Y, Oh W, Im I (2023) The voice of commerce: How smart speakers reshape digital content consumption and preference. *Management Inform. Systems Quart.* 47(2):857–874.

Sun C (2025) How does prepopulating search bars with keywords affect online consumer behavior? A field experiment. *Marketing Sci.* Forthcoming.

Sun C, Adamopoulos P, Ghose A, Luo X (2022) Predicting stages in omnichannel path to purchase: A deep learning model. *Inform. Systems Res.* 33(2):429–445.

Tan TF, Netessine S (2020) At your service on the table: Impact of tabletop technology on restaurant performance. *Management Sci.* 66(10):4496–4515.

Xu Y, Ghose A, Xiao B (2023) Mobile payment adoption: An empirical investigation of Alipay. *Inform. Systems Res.* 35(2):807–828.

Xu K, Chan J, Ghose A, Han SP (2017) Battle of the channels: The impact of tablets on digital commerce. *Management Sci.* 63(5):1469–1492.

Xu J, Forman C, Kim JB, Van Ittersum K (2014) News media channels: Complements or substitutes? Evidence from mobile phone usage. *J. Marketing* 78(4):97–112.

Yeverechyahu D, Mayya R, Oestreicher-Singer G (2024) The impact of Large Language Models on open-source innovation: Evidence from GitHub Copilot. Preprint, submitted September 16, http://dx.doi.org/10.2139/ssrn.4684662.

Zhang X, Goh KY, Sun C, Zhang R (2024) The value of AI-generated metadata for UGC platforms: Evidence from a large-scale field experiment. Working paper, National University of Singapore, Singapore.

Zhou M, Geng D, Abhishek V, Li B (2020) When the Bank comes to you: Branch network and customer Omnichannel banking behavior. *Inform. Systems Res.* 31(1):176–197.